DA6300 — Quantum Computing and Machine Learning

Tier	C
Phase (order)	7
Do this first	DA1001
Estimated effort	12 weeks
Unlocks	Research

Roadmap

1. Qubits, gates, circuits; variational algorithms; quantum ML.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Introduction to Quantum Computing and Quantum Hardware (38 videos). Qiskit *Introduction to Quantum Computing and Quantum Hardware* (official IBM, 2020 summer school): with labs.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Quantum Machine Learning | 2021 Qiskit Global Summer School (25 videos). Qiskit *Quantum Machine Learning* (2021 global summer school, official): the QML half.
- Quantum Algorithms and Cryptography | IIT Madras (59 videos). NPTEL *Quantum Algorithms and Cryptography* (IIT Madras).
- Introduction to Quantum Computation (33 videos). IISc *Introduction to Quantum Computation*.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–4	Qubits, states, circuits, measurement. Install Qiskit; read the matching chapters of the IBM Quantum Learning course.
2	5–7	Writing and running quantum programs. Implement Deutsch-Jozsa and Grover.
3	8–13	Quantum Fourier transform and Shor's algorithm.
4	14–16	Quantum error correction with repetition codes.
5	17–22	Superconducting qubits, circuit QED (hardware; optional).

Week	Lectures	What to do
6	23–28	Quantum chemistry and VQE. Then the quantum-ML series (alt) for variational classifiers.

▼ **All lectures in *Introduction to Quantum Computing and Quantum Hardware* (38 videos)**

1. Qiskit Global Summer School 2020 — The Largest Global Quantum Computing Summer School
2. 1. Qubits and Quantum States, Quantum Circuits, Measurements - Part 1
3. 2. Qubits and Quantum States, Quantum Circuits, Measurements - Part 2
4. 3. Qubits and Quantum States, Quantum Circuits, Measurements - Part 3
5. 4. Writing and Running Quantum Programs - Part 1
6. 5. Writing and Running Quantum Programs - Part 2
7. 6. Writing and Running Quantum Programs - Part 3
8. 7. Shor's Algorithm I: Understanding Quantum Fourier Transform, Quantum Phase Estimation - Part 1
9. 8.Shor's Algorithm I: Understanding Quantum Fourier Transform, Quantum Phase Estimation - Part 2
10. 9. Shor's Algorithm I: Understanding Quantum Fourier Transform, Quantum Phase Estimation - Part 3
11. 10. Shor's algorithm II: From Factoring to Period-Finding, Writing the Quantum Program - Part 1
12. 11. Shor's algorithm II: From Factoring to Period-Finding, Writing the Quantum Program - Part 2
13. 12. Shor's algorithm II: From Factoring to Period-Finding, Writing the Quantum Program - Part 3
14. 13. Quantum Error Correction using Repetition Codes - Part 1
15. 14. Quantum Error Correction using Repetition Codes - Part 2
16. 15. Quantum Error Correction using Repetition Codes - Part 3
17. 16. Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions - Part 1
18. 17. Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions - Part 2
19. 18. Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions - Part 3
20. 19. Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Pt 1
21. 20. Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Pt 2
22. 21. Superconducting Qubits II: Circuit Quantum Electrodynamics, Readout and Calibration Methods Pt 3
23. 22. Quantum Chemistry I: Obtaining the Qubit Hamiltonian for H2 and LiH - Part 1
24. 23. Quantum Chemistry I: Obtaining the Qubit Hamiltonian for H2 and LiH - Part 2
25. 24. Quantum Chemistry I: Obtaining the Qubit Hamiltonian for H2 and LiH - Part 3
26. 25. Quantum Chemistry II: Finding the Ground States of H2 and LiH - Part 1
27. 26. Quantum Chemistry II: Finding the Ground States of H2 and LiH - Part 2
28. 27. Quantum Chemistry II: Finding the Ground States of H2 and LiH - Part 3
29. How to Get a Job in Quantum Computing, Career Panel
30. Introduction to Quantum Computing and Quantum Hardware — Lab 1
31. Introduction to Quantum Computing and Quantum Hardware — Lab 2
32. Introduction to Quantum Computing and Quantum Hardware — Lab 3
33. Introduction to Quantum Computing and Quantum Hardware — Lab 4

34. Introduction to Quantum Computing and Quantum Hardware — Lab 5
35. Introduction to Quantum Computing and Quantum Hardware — Lab 6
36. Introduction to Quantum Computing and Quantum Hardware — Lab 7
37. Introduction to Quantum Computing and Quantum Hardware — Lab 8
38. Introduction to Quantum Computing and Quantum Hardware — Lab 9

Books, notes and websites

- IBM Quantum Learning (the course's resource); Nielsen & Chuang

Practice / proof of understanding

Run Grover and a VQE on a simulator.

What your IITM course sheet says

From the PDF: DA6300

Official description: Quantum computing hardware has matured to the extent that there are some scientific experiments for which results using the currently available hardware outperform the results from using state-of-art classical supercomputers. On the other hand, machine learning (especially deep learning) models have been increasingly successful in a variety of domains spanning across science, economics and engineering. This course aspires to cover this interplay between quantum computing and machine learning from both the directions: 1. Quantum ML: Machine learning models built with quantum algorithms which leverage quantum computing principles 2. ML for Quantum: Use of (classical) machine learning in enhancing the quantum computing software stack

Official syllabus (trimmed):

> 1. Quantum computing foundations. (3 weeks) Qubits – Gates – Relevant quantum mechanics foundations – Circuit identities – Phase kickback – Quantum circuits – Teleportation demo - Dynamic Circuits - Transpilation - Error mitigation
> 2. Variational quantum algorithms and applications. (3 weeks) Optimization problems - Quantum Encoding schemes (Basis, Angle, High dimensional, Amplitude, QRACs) - Eigenvalue problem and Quantum phase estimation algorithms - Variational eigenvalue solver - Examples of evaluating ground state energy - Other example applications in finance, chemistry and biosciences.
> 3. Quantum Machine Learning (5 weeks) Variational Algorithms for machine learning - Quantum-classical hy- brid learning algorithms - Quantum Neural Networks(QNN) - Variational Quantum Algorithms (VQA) - Quantum Kernel Estimation (QKE) - Quantum Support Vector Machine (QSVM) - Quantum Generative Adversarial Net- work (QGANs) - Quantum Convolutional Network (QCNN) - Quantum Graph Convolutional Network (QGCNN) - Quantum Recurrent Neural Networks QRNN) - Quantum Autoencoders, Quantum Boltzmann Machines (QBM).
> 4. Machine learning applications to quantum computing stack (3 weeks) ML for compilation - ML for error correction - ML for error mitigation - ML for quantum code assistant

Official textbooks: 1. Nielsen and Chuang, Quantum Computation and Quantum Information, Cambridge University Press 2. Cerezo et al., Variational quantum algorithms, Nature Reviews Physics 3. Simeone, An Introduction to Quantum Machine Learning for Engineers, Foundations and Trends in Signal Processing 4.

IBM Quantum Learning, https://learning.quantum.ibm.com/ 5. Relevant research papers in the topics of interest.

Official references: 1. Nielsen and Chuang, Quantum Computation and Quantum Information, Cambridge University Press 2. Cerezo et al., Variational quantum algorithms, Nature Reviews Physics 3. Simeone, An Introduction to Quantum Machine Learning for Engineers, Foundations and Trends in Signal Processing 4. IBM Quantum Learning, https://learning.quantum.ibm.com/ 5. Relevant research papers in the topics of interest